MORONI: Polygon-Guided Cellular Textures via Roots-of-Unity Polygons and Phase-Modulated Boundaries

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ABSTRACT

Voronoi and Worley noise are foundational cellular primitives in procedural graphics and games, but standard Euclidean assignment always yields rounded, convex cells with limited art direction. Game artists and technical directors cannot directly prescribe target polygonal silhouettes (such as masonry stonework, ceramic tiles, or cracked mud) or control inter-cell crack dynamics.

This paper presents **MORONI**, a practical cellular texture engine that shapes procedural cells using roots-of-unity regular polygons and phase-modulated boundaries. We assign pixels through a lightweight three-term energy combining center distance, polygon signed distance, and angular harmonic phase waves, followed by an adjustable margin threshold for grout and mortar release. We demonstrate four concrete properties of the technique: (1) **Shape Control:** polygon clipping frees cells from space-filling constraints, raising target silhouette IoU from 0.551 to 0.939, while MORONI's signed energy arbitrates contested borders to reach 0.954 IoU. (2) **Search Exactness:** an enclosing-disk lower bound accelerates candidate evaluation with zero label disagreements against brute-force search across all 737,280 tested pixels (120 test cases). (3) **Lookup Precision:** a compact 72×72 bilinear lookup table reproduces polygon distance with 2.15×10^{-5} mean absolute error and only 0.00625% label mismatch against analytical geometry. (4) **Real-Time GPU Parity:** an interactive WebGL2 studio evaluates the identical mathematical field at 60 FPS on consumer hardware, exporting synchronized albedo, normal, height, and roughness maps.

The interactive studio and open-source implementation are available at https://pikachuxxxx.github.io/Moroni-Studio/ and https://github.com/Pikachuxxxx/Moroni-Studio.

KEYWORDS

procedural texture, cellular noise, Voronoi diagram, signed distance field, roots of unity, phase modulation, real-time rendering, WebGL

1 INTRODUCTION

Cellular noise is foundational to procedural visual effects, game rendering, and virtual worlds. Natural and architectural surfaces — cracked desert mud, masonry stone walls, ceramic floor tiles, shattered glass, reptilian scales, and weathered paint — exhibit characteristic cellular tiling structures. In modern real-time pipelines (Unreal Engine Material Graph, Unity Shader Graph, Blender Cycles/EEVEE, and Substance 3D Designer), Voronoi and Worley noise [22, 23] are widely used for their compact representations and efficient evaluation.

Despite their ubiquity, point-site cellular primitives couple site placement and cell boundaries. For classical Euclidean Voronoi diagrams, boundaries lie on perpendicular bisectors. A square lattice can produce square cells; prescribed arbitrary targets cannot generally be controlled independently while retaining a full partition. Domain warping provides a useful alternative but changes the coordinate frame and hence the targets as well as their boundaries.

Weighted diagram generalizations relax the perpendicular bisector constraint, but remain structurally constrained. Power (Laguerre) diagrams [1] replace Euclidean distance $\|\mathbf{x} - \mathbf{u}_c\|$ with $\|\mathbf{x} - \mathbf{u}_c\|^2 - w_c$, and additively weighted (Apollonius) diagrams replace it with $\|\mathbf{x} - \mathbf{u}_c\| - w_c$. However, power diagram cells are intersections of affine half-planes and are mathematically guaranteed to be *convex*; Apollonius cells are guaranteed to be *connected*. Concavity is excluded by power diagrams, while nonempty additively weighted cells are connected. Raster measurements alone do not prove that a continuous MORONI cell violates either property.

Two Geometric Challenges in Polygon-Guided Cellular Synthesis. Bridging the gap between procedural efficiency and art directability requires attaching explicit target polygons to cells. However, constructing an energy field over polygon cells introduces two distinct geometric challenges:

(1) **The Boundary Distance Dilemma**: An unsigned distance $\text{dist}(\mathbf{x}, \partial P_c)$ achieves its minimum *on* the boundary outline ∂P_c. Consequently, penalizing boundary distance favors the perimeter over deeper interior points. A signed field favors interior points over the perimeter when this term is minimized (§3.2). This preference does not guarantee recovery of a target silhouette in a competing field.
(2) **Space-Filling Constraint vs. Target Silhouettes**: Under a strict space-filling arg min partition, every pixel must be assigned to some cell, leaving no unassigned space. For jittered sites, adjacent cells compete along their common borders, distorting target shapes (such as triangles or quads) into irregular polygons with compromise edges. Recovering clean target silhouettes requires decoupling cell assignment from strict space filling. Introducing a coverage margin m releases pixels whose distance exceeds a threshold, allowing cells to preserve their prescribed silhouettes while simultaneously providing a monotonic coverage dial (§3.4).

Summary of Contributions.

- A polygon-guided cellular texture recipe combining center distance, signed polygon distance, and angular phase modulation (§3).
- Matched clipping and phase ablations over eight seeds, exposing both the modest assignment benefit and the fidelity cost of phase (§4).
- Raster solidity and component measurements with explicit sampling, connectivity and filtering conventions (§5).
- Conservative candidate bounds, finite adaptive search, and a lookup error analysis distinguishing approximation from acceleration (§7).

- A WebGL2 studio using shared geometry, actual GPU timer queries, cross-implementation tests and explicitly defined map exports (§8).
- Full generator algorithms and a comparative operational matrix (Appendix A).

These contributions combine established primitives; roots of unity and signed distance are not claimed as new, and no new classification of weighted Voronoi diagrams is asserted.

2 RELATED WORK

Procedural pattern synthesis in computer graphics spans gradient noise, cellular diagrams, distance fields, and physical simulation.

2.1 Procedural Noise Functions

Procedural noise began with Perlin's gradient noise [18], followed by multi-scale spectral representations such as wavelet noise by Cook and DeRose [2]. Lagae et al. [11] introduced Gabor noise, enabling exact spectral control via sparse convolution. The comprehensive survey by Lagae et al. [10] reviews noise functions for procedural texturing; modern rendering systems combine such functions with material models [19]. While gradient and Gabor noise excel at continuous, band-limited phenomena (clouds, marble, wood grain), piecewise cellular assignment is a different primitive; thresholds and domain warps can combine these families.

2.2 Cellular Diagrams and Weighted Extensions

Cellular noise descends from Georgy Voronoi's 1908 treatise on quadratic forms [22]. Steven Fortune [5] pioneered optimal sweepline geometry for planar Voronoi diagrams. Lloyd [12] introduced iterative centroidal Voronoi relaxation, widely adopted in graphics for stippling and remeshing. Steven Worley [23] revolutionized shader texturing by introducing the cellular basis function $F_k(\mathbf{x})$, measuring distances to the k-th nearest feature points on a spatial grid. McEwan et al. [13] discuss efficient computational noise in GLSL; their work is not evidence for a particular MORONI implementation.

To provide size control, weighted Voronoi diagrams were introduced. In Power diagrams (Laguerre geometry) [1], each site carries a scalar weight w_c subtracting from squared distance: $\|\mathbf{x}-\mathbf{u}_c\|^2-w_c$. Power diagrams retain affine linear bisectors and have convex cells. In additively weighted (Apollonius) diagrams, weights subtract directly from Euclidean distance: $\|\mathbf{x}-\mathbf{u}_c\|-w_c$, producing hyperbolic curved boundaries. Nonempty additively weighted cells are connected [8], but may be nonconvex. Section 5 separates these continuous properties from thresholded raster observations.

2.3 Procedural Fracture vs. Physical Simulation

Physical fracture modeling accurately simulates crack propagation using finite element methods (FEM) or boundary element methods (BEM). O'Brien and Hodgins [17] established continuum mechanics fracture for graphics. Desbenoit et al. [3] modeled surface cracks and peeling patterns, and Nealen et al. [16] surveyed physically based deformable models. These methods address physical deformation and fracture. Their costs depend on the model and discretization; we do not provide a timing comparison with them or claim equivalent physical accuracy.

Procedural crack generation offers a lightweight alternative. Mould [14] synthesized image-guided surface crack patterns using shortest-path graph propagation on lattices. Such graph constructions offer a different control model. MORONI evaluates a finite field independently per pixel, using uploaded polygon data; its phase term supplies crack-like stylization, not physical fracture propagation.

2.4 Implicit Surfaces and Distance Fields

Distance transforms map discrete masks to continuous distance fields [4]. Hart [7] demonstrated sphere tracing against implicit signed distance fields (SDF). In real-time demoscene and shader art, signed distance functions enable compact procedural modeling. MORONI bridges signed distance fields with cellular partition theory, using signed boundary fields to influence assignment, followed by optional clipping. Neither term alone guarantees a target silhouette.

2.5 Controllability in Procedural Content Generation

Controllability is the central challenge in procedural content generation (PCG). Smelik et al. [20] and Vanegas et al. [21] emphasize that procedural algorithms must provide intuitive, declarative handles rather than opaque mathematical parameters. Müller et al. [15] demonstrated rule-based grammar derivation for procedural architecture, providing an example of structured procedural control. MORONI exposes shape, size, irregularity, coverage, and phase controls, but these interact and their usability has not been evaluated in an artist study.

Guehl et al. [6] combine point processes and rich kernels in Point Process Texture Basis Functions. MORONI addresses the narrower case of explicit finite polygon targets. No superiority in coverage, image matching, or perceptual quality over PPTBF is claimed.

3 METHOD

3.1 Cell Parameterization

Each cell $c \in \{0,\dots,N-1\}$ owns a center $\mathbf{u}_c$ and an explicit n_c-gon P_c whose vertices are generated as jittered roots of unity:

$$z_{c,k} = r_c \exp\Big(i\big(\tfrac{2\pi k}{n_c} + \delta_{c,k}\big)\Big), \quad \delta_{c,k} \sim \mathcal{U}(-J_c, J_c), \tag{1}$$

for $k = 0,\dots,n_c-1$, offset by cell center $\mathbf{u}_c$. Centers are positioned on a jittered Cartesian lattice of spacing s:

$$\mathbf{u}_c = \begin{pmatrix} c_x \cdot s + \xi_{c,x} \\ c_y \cdot s + \xi_{c,y} \end{pmatrix}, \quad \xi_{c,x}, \xi_{c,y} \sim \mathcal{U}(-0.13s, 0.13s). \tag{2}$$

Here, n_c governs target cell shape (e.g., 3 for triangles, 4 for quads), $r_c = r(0.9 + 0.2\xi_c)$ with $\xi_c \sim \mathcal{U}(0,1)$ governs size, and $J_c = \min(J, 0.45\pi/n_c)$ limits angular irregularity. This cap preserves circular vertex order and angular gaps below π for $n_c \geq 3$, giving simple convex input polygons containing their centers. The assigned regions need not be convex. Python and JavaScript share a Mulberry32 sequence and draw order; the resulting realization differs from earlier prototypes. The supported field is finite, with default 8×12 sites and $s = .9$. When unspecified, n_c is sampled uniformly from $\{5, 6, 7, 8\}$. The shader accepts at most 256 cells with 3–12 vertices.

Samples are pixel centers over $[-s/2, (C-.5)s] \times [-s/2, (R-.5)s]$, with image y downward.

3.2 Signed Boundary Field

Let P_c denote the polygon of Eq. 1. We define the signed boundary distance field $s_c(\mathbf{x})$:

$$s_c(\mathbf{x}) = \begin{cases} -\operatorname{dist}(\mathbf{x}, \partial P_c) & \mathbf{x} \in P_c \\ +\operatorname{dist}(\mathbf{x}, \partial P_c) & \text{otherwise,} \end{cases} \tag{3}$$

where $\operatorname{dist}(\mathbf{x}, \partial P_c) = \min_{\mathbf{y} \in \partial P_c} \|\mathbf{x} - \mathbf{y}\|_2$.

Mathematical Rationale for the Signed Field. Consider an energy term based on unsigned distance $\beta \operatorname{dist}(\mathbf{x}, \partial P_c)$. Because dist ≥ 0 with $\operatorname{dist}(\mathbf{x}, \partial P_c) = 0$ precisely on the perimeter ∂P_c, the term attains its minimum along the boundary. An unsigned boundary-distance term favors the perimeter rather than deeper interior points. The signed term instead has negative values inside the polygon, attaining its most negative value at a center of a largest inscribed disk. Its behavior is not monotonic along every path toward the medial axis, and competition with the other terms can still distort or remove a cell. Figure 1 shows the separate effect of releasing coverage after assignment.

3.3 The Three-Term Energy Formulation

Let $\mathbf{d} = \mathbf{x} - \mathbf{u}_c$, $d_c = \|\mathbf{d}\|_2$, and $\theta_c = \operatorname{atan2}(d_y, d_x)$. We assign each pixel $\mathbf{x}$ to the cell minimizing:

$$E_c(\mathbf{x}) = \underbrace{\alpha d_c}_{\text{size}} + \underbrace{\beta\, s_c(\mathbf{x})}_{\text{shape}} + \underbrace{\gamma\big(1 - \cos(\kappa\theta_c + \lambda d_c)\big)}_{\text{fracture}}, \tag{4}$$

$$L(\mathbf{x}) = \arg \min_{c \in \{0,\ldots,N-1\}} E_c(\mathbf{x}). \tag{5}$$

Default weights are $\alpha = 0.58$, $\beta = 0.32$, $\gamma = 0.10$, $\kappa = 2$, and $\lambda = 1$. Setting $\beta = \gamma = 0$ collapses $E_c(\mathbf{x})$ to αd_c, recovering nearest-site assignment for $\alpha > 0$. Weights and λ must be finite and nonnegative, with integer $\kappa \in \{1, \ldots, 6\}$. Ties select the smallest index; all-zero weights select cell zero. For $d_c < 10^{-6}$ the phase contribution is set to zero. There is no direction-independent phase limit at the site; this is an explicit convention, not continuity of atan2.

The phase term $\gamma(1 - \cos(\kappa\theta_c + \lambda d_c))$ acts as an angular standing wave propagating radially outward. Because neighbouring cells have independent centres, their respective polar angle frames θ_i, θ_j differ in contested border regions, causing cell boundaries to deviate from straight bisectors into intricate, wandering crack paths.

3.4 Coverage Margin and Space-Filling Decoupling

Equation 5 assigns every pixel, producing a space-filling partition of the image domain. Under point-site Voronoi assignment with jittered positions, cell boundaries are formed by perpendicular bisectors that generically intersect at 3-valent junctions; as a result, cells cannot conform to prescribed target silhouettes (such as equilateral triangles or rectangles) without distorting neighboring boundaries. Even when introducing polygon distance into the energy, a strict space-filling partition forces adjacent cells to share all intervening boundaries, producing compromise shapes (the current default-phase triangle experiment gives SF = 0.556).

Table 1: Shape fidelity (mean IoU against the target polygon), averaged over eight seeds at 320×200. Both methods use the same polygon clipping at $m = 0$; MORONI uses default $\gamma = .1$. Clipping is the dominant source of fidelity.

n_c	Voronoi	Vor.+Clip	MORONI	MOR.+Clip
3	0.551	0.939	0.556	0.945
4	0.701	0.858	0.692	0.851
5	0.775	0.839	0.751	0.822
6	0.761	0.795	0.732	0.776
8	0.736	0.752	0.709	0.734

To restore shape control and allow cells to reflect target silhouettes, we decouple cell assignment from the space-filling requirement by introducing a coverage margin $m \in \mathbb{R}$:

$$L_m(\mathbf{x}) = \begin{cases} \varnothing & s_{L(\mathbf{x})}(\mathbf{x}) > m \\ L(\mathbf{x}) & \text{otherwise.} \end{cases} \tag{6}$$

Pixels whose signed distance to their assigned polygon exceeds m are released as background / grout ($\varnothing$). This accomplishes two critical objectives:

(1) It frees cells from the space-filling constraint, permitting them to adhere closely to their prescribed silhouettes ($m = 0.0$ clips cells cleanly to their polygon boundaries, giving mean triangular shape fidelity 0.945 in Table 1).
(2) It provides a monotonic coverage threshold: $m > 0$ permits controlled cell dilation, $m = 0$ clips cells exactly to their polygons, and $m < 0$ erodes them. Coverage is monotonic for fixed labels, but physical gap width is not guaranteed constant; released pixels are not reassigned to another cell.

4 DIRECTABILITY AND CONTROL QUANTITATIVE STUDIES

Metric. We evaluate shape fidelity (SF) as the mean Intersection over Union (IoU) between each rendered cell and its generating polygon:

$$\mathrm{SF} = \frac{1}{|C_{\text{valid}}|} \sum_{c \in C_{\text{valid}}} \frac{|\{\mathbf{x} : L_m(\mathbf{x}) = c\} \cap P_c|}{|\{\mathbf{x} : L_m(\mathbf{x}) = c\} \cup P_c|}. \tag{7}$$

Eligibility uses the target polygon intersected with the image domain, not the output area: targets with at least 16 pixels are included, and vanished output cells score zero. We average cells within each seed and then eight seeds (7, 22, 42, 100–104). Unless stated otherwise, this study uses 320×200, 96 sites, $r = .62$, $J = .05$, analytical distance and default energy weights. Per-seed values, standard deviations and coverage are saved.

Table 1 isolates clipping from assignment. For triangles, clipped Voronoi reaches 0.939 and default MORONI reaches 0.945; for the other side counts in this experiment, default phase reduces IoU relative to clipped Voronoi. In the no-phase ablation, MORONI reaches 0.954 for triangles, with a paired improvement of 0.01474 ± 0.00069 over clipped Voronoi (mean and sample standard deviation across seeds). The corresponding improvements for 4, 5, 6 and 8 sides are 0.00301, 0.00495, 0.00174 and 0.00072. Phase trades silhouette fidelity for variation rather than optimizing fidelity. The artifact

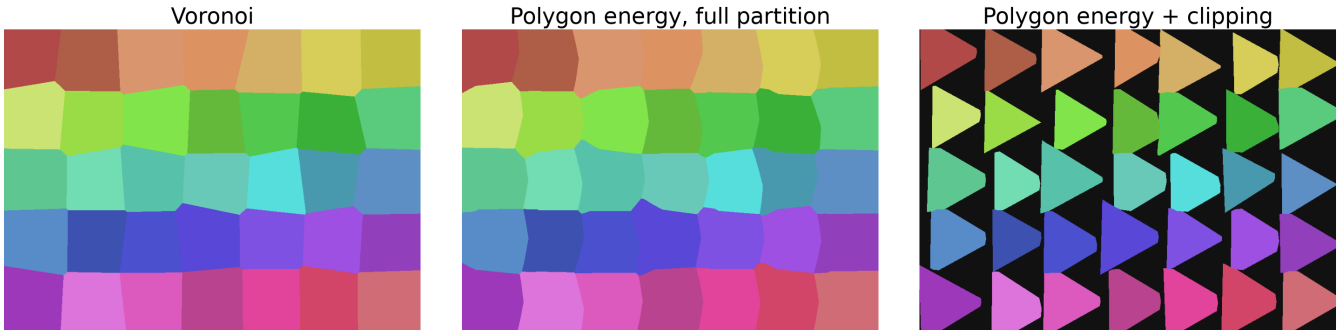


Figure 1: Vertex count $n_c = 3$, seed 7, 5×7 sites, $r = .62$, $J = .05$, and $\gamma = 0$ throughout. Voronoi (left) uses nearest sites. A strict space-filling partition (centre) forces shared boundaries between adjacent cells. The signed field with coverage margin m (right) retains portions of triangle targets; overlaps still require arbitration.

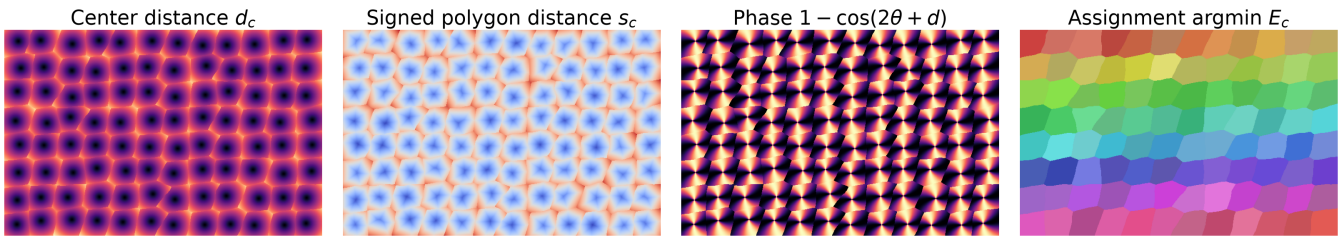


Figure 2: The three component fields of Eq. 4, evaluated for the winning cell at each pixel, and the resulting labelling. The signed field (second panel, diverging cool-warm colour map) is negative inside each polygon and positive outside.

Table 2: Cell coverage fraction is monotonic in margin m, confirming set inclusion at fixed assignment. Means use the same eight seeds, default field geometry ($r = .5, J = .35$), default weights and analytical distance; this does not measure constant physical mortar width.

Margin m	-0.10	-0.05	0.00	0.05	0.12	0.25
Coverage (%)	47.0	61.2	74.6	85.1	94.2	99.6

also includes a fixed sinusoidal domain-warp baseline with matched clipping; it is an illustrative alternative, not a tuned perceptual comparison. Table 2 reports increasing coverage from 47.0% to 99.6% as the threshold increases; coverage and gap width should not be conflated.

5 RASTER GEOMETRY AND WEIGHTED-DIAGRAM COMPARISONS

Continuous Properties and Raster Evidence.

- **Power Diagrams**: In a power diagram, cell boundaries are lines

$$\mathbf{x} \cdot (\mathbf{u}_i - \mathbf{u}_j) = \tfrac{1}{2}\big(\|\mathbf{u}_i\|^2 - \|\mathbf{u}_j\|^2 - w_i + w_j\big). \tag{8}$$

  Cells are intersections of half-planes, which are mathematically guaranteed to be *convex* in the continuous plane $\mathbb{R}^2$ [1].

- **Apollonius Diagrams**: In additively weighted diagrams, bisectors are branches of hyperbolas. While cells can be non-convex, nonempty cells are mathematically guaranteed to be *connected*. A continuous partition containing a disconnected cell (multiple disjoint spatial components) cannot be produced by any Apollonius diagram.

Experimental Benchmark. We evaluate solidity $S = A_{\text{pix}}/A_{\text{hull}}$, the ratio of pixel area to convex-hull area, and components on 320×200 raster labels, seed 22, without clipping. Hulls use all four corners of each pixel. Components use 8-connectivity and minimum area $\lceil 10^{-4} WH \rceil$ (7 pixels here). Cells below this area are excluded. The solidity threshold $S < .98$ is descriptive. Baselines share sites; power weights are r_c^2 and Apollonius weights are r_c. Analytical distance is used for MORONI.

Table 3 describes the sampled progression. Euclidean Voronoi and power cells are convex in the plane, yet their raster solidity can be below one. At $\gamma = .35$, this MORONI sample has median solidity 0.765 and 8.3% of cells with multiple retained components. Changing resolution, connectivity, threshold or clipping changes those statistics. The artifact also evaluates 320×200, 640×400 and 960×600 under both 4- and 8-connectivity, with and without clipping. These results describe stylized raster regions, not exclusion from every weighted-diagram family or physical fracture behavior.

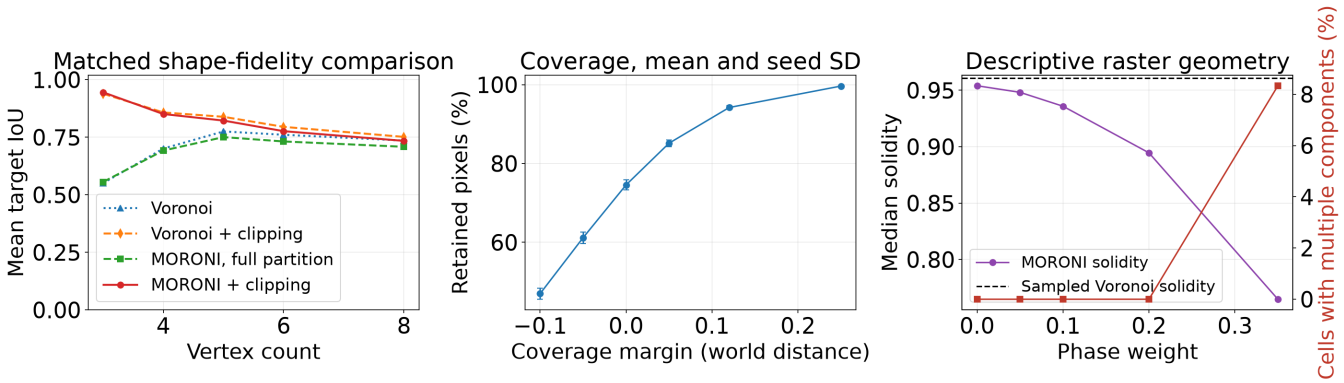


Figure 3: Quantitative control behavior. Left: Shape fidelity against vertex count n_c. Centre: Coverage fraction against margin m. Right: Cell solidity and multiple-component percentage against phase weight γ, under the stated raster convention. These measurements are not topology proofs.

Table 3: Raster metrics across procedural generators and phase weights at 320×200. These are raster statistics: $S < .98$ is not a continuous nonconvexity test, and multiple retained components are not a proof of continuous disconnection.

Generator	Solidity	$S < .98$ (%)	Mean comp.	Multiple comp. (%)
Euclidean	0.961	92.7	1.00	0.0
Power	0.960	95.8	1.00	0.0
Manhattan	0.942	96.9	1.00	0.0
Chebyshev	0.950	88.5	1.00	0.0
Apollonius	0.961	93.8	1.00	0.0
MORONI $\gamma = 0.00$	0.954	95.8	1.00	0.0
MORONI $\gamma = 0.05$	0.948	99.0	1.00	0.0
MORONI $\gamma = 0.10$	0.936	100.0	1.00	0.0
MORONI $\gamma = 0.20$	0.895	100.0	1.00	0.0
MORONI $\gamma = 0.35$	0.765	100.0	1.08	8.3

6 ARTIST CONTROLS AND MATERIAL PRESETS

Figures 4 and 5 demonstrate the five directable controls across illustrative presets (names indicate inspiration, not physical validation):

(1) **Cracked Mud**: $n_c = 7$, $r = 0.6$, $J = 0.40$, $\gamma = 0.30$, and $m = 0.05$. Produces stylized crack-like networks with irregular interlocking mud slabs.
(2) **Medieval Stone Wall**: $n_c = 5$, $r = 0.62$, $J = 0.12$, $\beta = 0.90$, $\gamma = 0.02$, and $m = 0.03$. Generates polygonal blocks with illustrative mortar recesses.
(3) **Ceramic Tile**: $n_c = 4$, $r = 0.60$, $J = 0.02$, $\beta = 1.20$, $\gamma = 0.00$, and $m = 0.01$. Yields quadrilateral tile patterns with gaps selected by the coverage margin.
(4) **Shattered Glass**: $n_c = 4$, $r = 0.70$, $J = 0.50$, $\gamma = 0.35$, $\kappa = 3.0$, $\lambda = 1.8$, and $m = 0.08$. Produces stylized sharp regions and irregular seams; no stress model is evaluated.

7 CONSERVATIVE ACCELERATION AND ERROR ANALYSIS

7.1 An Admissible Incumbent-Based Energy Bound

Evaluating brute-force energy over N cells for P pixels requires $O(NP)$ work. Each generating polygon is contained in its radius-r_c disk. Thus $s_c(\mathbf{x}) \geq d_c - r_c$: outside the polygon this follows from triangle inequality, and inside, distance to its boundary cannot exceed distance to the enclosing disk boundary. Nonnegative weights and phase yield

$$E_c(\mathbf{x}) \geq (\alpha + \beta)d_c - \beta r_c. \tag{9}$$

For hybrid distance, use $r_c + \varepsilon_c$, where ε_c bounds lookup error (Appendix C). Given incumbent U, reject only when

$$(\alpha + \beta)d_c - \beta(r_c + \varepsilon_c) > U. \tag{10}$$

Analytical mode uses $\varepsilon_c = 0$. Equality remains eligible for index ties. The real-arithmetic bound does not by itself prove floating-point correctness; the CPU uses a relative slack of 10^{-12} and separately tests observed agreement. No sampled polygon in-radius is treated as an exact bound.

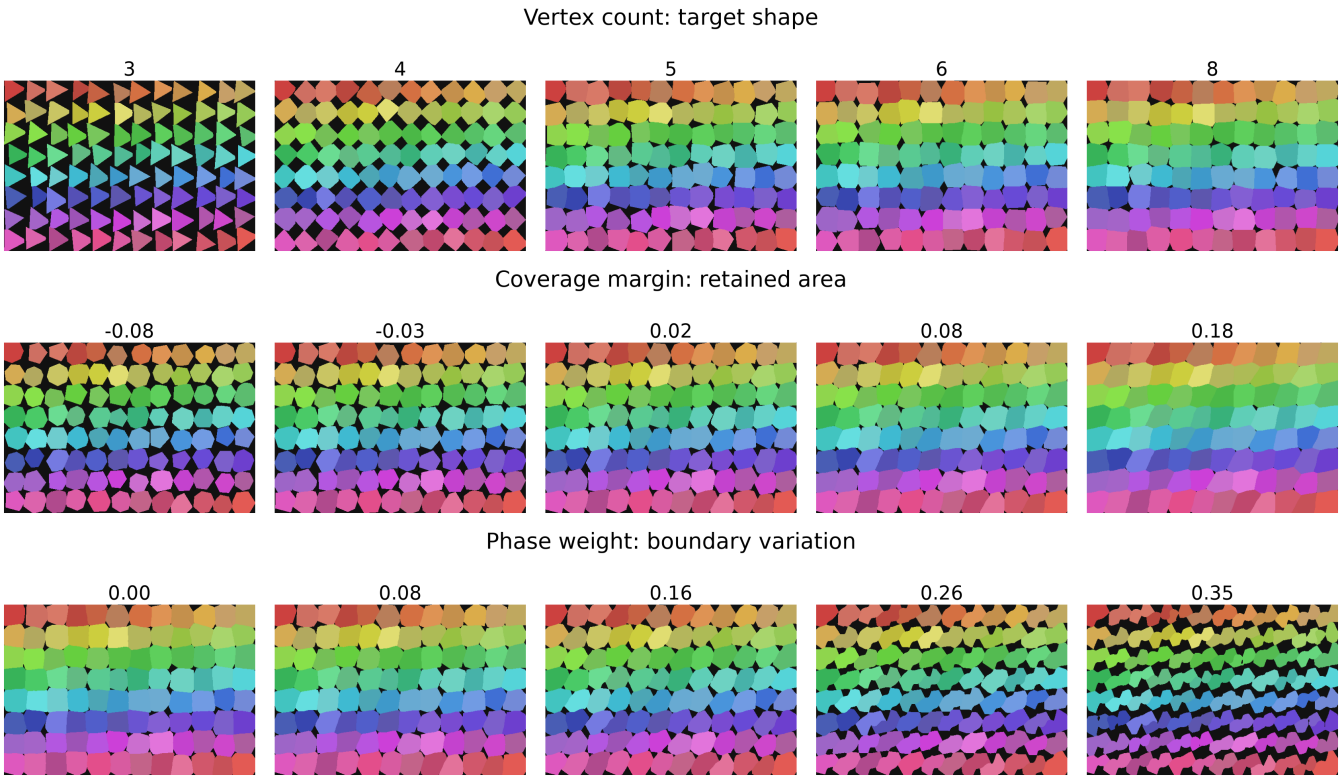


Figure 4: Separate exploration of three interacting controls: vertex count n_c (cell shape), coverage margin m (retained area), and phase weight γ (fracture intensity).

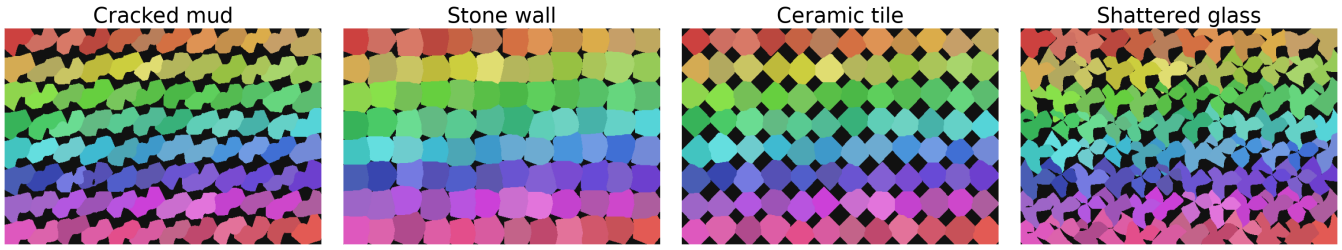


Figure 5: Illustrative material presets reached by tuning the controls: Cracked mud, Stone wall, Ceramic tile, and Shattered glass.

7.2 Lattice Binning and Ring Escalation

Cells reside on a jittered lattice of spacing s. The renderer visits Chebyshev rings around the nearest valid lattice anchor. Unvisited sites lie in up to four strips. Distance to half-planes containing these strips, expanded by $0.13s$ site jitter, lower-bounds the remaining site distance D. For example, the first unvisited right column j has $x_c \geq (j - .13)s$ and contributes $\max(0, (j - .13)s - x)$. Absent strips contribute $+\infty$; take the minimum over present strips. A query terminates only when

$$E_{\text{floor}} = (\alpha + \beta)D - \beta \max_c(r_c + \varepsilon_c) > U. \tag{11}$$

Otherwise rings continue until the finite field is exhausted. This also works for exterior queries and all-zero weights, without division by α or a fixed two-ring cutoff. Bins and bounds are recomputed when query positions or parameters change. The worst case remains $O(NP)$; no majority-at-ring-zero or constant-cost guarantee is made.

7.3 Phase Evaluation

The corrected implementation uses the direct expression in Eq. 4 for all supported harmonics 1–6; no phase LUT or trigonometric-elimination speedup is claimed. For $d_c < 10^{-6}$, the phase contribution is explicitly zero, since the angular term has no direction-independent limit at the center.

Algorithm 1 Adaptive Binned MORONI Renderer

Require: Query grid, finite field, nonnegative weights, optional margin
Recompute valid anchor bins for the current queries
$A \leftarrow$ all queries; $U \leftarrow \infty$; $L \leftarrow 0$
for each ring until field exhaustion or $A = \varnothing$ **do**
  for valid candidate c on this ring **do**
    Reject using Eq. 10, with comparison slack
    Evaluate remaining energies using the selected distance mode
    Update (U, L) by energy, then smallest index for ties
  end for
  Compute D from remaining lattice strips
  Remove queries satisfying Eq. 11
end for
Apply Eq. 6 only after assignment
return L

7.4 Discretization Error of the Signed Field LUT

In the CPU reference renderer, signed distance is tabulated over $[-B_c, B_c]^2$ with $B_c = 1.5r_c$. Queries inside use bilinear interpolation; queries outside use analytical polygon distance. A clamped lookup plus distance to the box is not an exact SDF extension. We test 4,000 uniform queries per polygon over $[-2B_c, 2B_c]^2$ for the 96 default polygons (query seed 42); approximately three quarters use the exact outside-box path. Label disagreement is measured on the default 320×200 field.

Table 4: Discretization error and label mismatch percentage of the bilinearly interpolated signed-field lookup table (LUT) against analytical point-to-polygon distance.

L	MAE	RMSE	Max. error	Labels (%)
36	0.000089	0.000575	0.02096	0.0234
72	0.000022	0.000191	0.00847	0.0063
144	0.000005	0.000064	0.00410	0.0016

As documented in Table 4, the 72×72 hybrid lookup has MAE 2.15×10^{-5}, RMSE 1.91×10^{-4} and maximum observed error 0.00847 over this distribution. Four of 64,000 default-field labels differ from analytical assignment (0.00625%). These are spatial distance errors, not angular precision. The interpolation error bound and sufficient winning energy lead are derived in Appendix C; no universal 10^{-5} near-tie threshold or mismatch rate is asserted.

8 REAL-TIME GPU SYNTHESIS AND INTERACTIVE STUDIO

Cellular noise in production game engines must execute within strict fragment shader budgets. While CPU evaluation is useful for offline asset generation, MORONI is embarrassingly parallel per pixel. To demonstrate real-time feasibility, facilitate rigorous peer review, and provide an interactive demonstration, we developed **MORONI Studio**, a zero-dependency WebGL2 companion application hosted online:

Interactive Online Studio & Open-Source Repository

Live WebGL2 Studio:
https://pikachuxxxx.github.io/Moroni-Studio/

GitHub Repository:
https://github.com/Pikachuxxxx/Moroni-Studio

A zero-dependency, open-source WebGL2 studio for interactive control, matched-site A/B comparison, energy inspection and PNG map export. SVG exports input polygons, not final assigned-cell contours. Reference implementations, tests and raw measurements are MIT licensed.

Table 5: GPU draw execution on AMD Radeon RX 9070, ANGLE/D3D11, Windows. Each condition uses 12 warmups and 40 measured timer queries; single-sample pixels. JavaScript, upload, readback and encoding are excluded.

Pixels	Output	Draws	Median ms
640×400	Unlit color	40	0.246
1280×800	Unlit color	40	0.807
1920×1080	Unlit color	40	1.604
640×400	Normal	40	1.180

In the WebGL2 fragment shader implementation, JavaScript uploads the same generated sites, polygon vertices and identity palette used by its CPU renderer. The shader traverses the complete finite list, rejecting candidates with the disk bound, and uses analytical segment distance and direct phase. There is no fixed hashed 5×5 neighborhood. GPU timer queries measure execution asynchronously and discard disjoint intervals [9]; host submission time is not called GPU time. Results in Table 5 describe the stated device and are not universal FPS or CPU/GPU speedup claims.

The studio retains interactive controls, A/B inspection, energy views and PNG exports up to 2048 pixels wide with preserved lattice aspect ratio. Paper Colors and albedo are unlit; shaded mode is removed. The HUD samples GPU labels at 180×120 and reports descriptive solidity/components, not diagram-family classification. Four-sample preview averaging is optional; exports and metric labels use single samples. Export semantics and map conventions are specified in Appendix D.

8.1 Code Availability and Reproducible Checks

To enable reproduction, code and experimental pipelines are available under the MIT license at https://github.com/Pikachuxxxx/Moroni-Studio.

- **CPU source:** src/moroni.py, with wrapper moroni.py, implements analytical/hybrid distance and adaptive bounded search.
- **GPU source:** app/moroni-glsl.js evaluates uploaded polygons and phase, with optional timer-query measurements.
- **Experiments:** The scripts in benchmarks/ save raw results, refresh the existing tables, and render the figures.
- **Checks:** Python and Node regression tests, plus app/validation.html, compare generated geometry and labels.

Accelerated and full CPU searches agree on 737,280 pixels in 120 cases, including both distance modes, changed parameters, harmonics and exterior query domains. Nine browser cases contain

Table 6: Median CPU runtime (ms) for matching hybrid distance implementations and a KD-tree Voronoi baseline. Ratios are computed from those medians.

Pixels	Voronoi	MORONI	Full	Full/M. / Vor./M.
320×200	11.8	67.7	765.5	11.3× / 0.17×
640×400	46.5	188.7	6964.8	36.9× / 0.25×
960×600	106.1	379.4	16283.5	42.9× / 0.28×

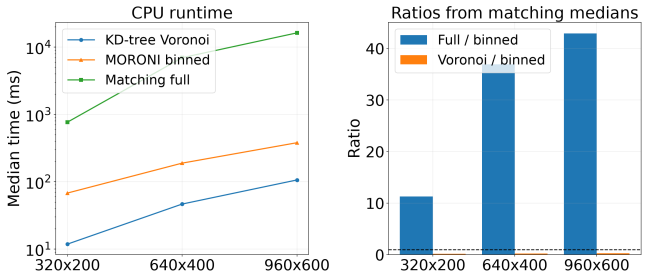


Figure 6: CPU runtime and relative performance.

55,296 pixels with no GPU pruned/full differences; GPU versus CPU differs at one pixel whose winning energy lead is approximately 2.61×10^{-6}. The default Python/JS fixture agrees. These sampled tests do not prove exhaustive floating-point agreement. Exact commands and environments are in the reproduction guide.

9 CPU PERFORMANCE BENCHMARKS

Benchmarking Protocol. Windows, AMD Ryzen 9 7900X, NumPy 2.3.5, SciPy 1.18.1 and Python 3.12; no JIT. The default 96-cell field uses seed 22. One warmup precedes five interleaved rounds; medians and all samples are saved. Cell/LUT construction is excluded; KD-tree construction and query are included with `workers=1`. MORONI and its full reference use the same 72×72 hybrid distance. Analytical timings are also in the artifact.

The accelerated hybrid renderer is 11.3–42.9 times faster than its matching full search on these sizes, while slower than KD-tree Voronoi. Full hybrid search performs analytical distance for out-of-box candidates, so its cost is substantial. The ratio depends on the implementation, parameters and image size; it is not an algorithm-independent advantage. No CPU/GPU ratio is reported because precision, execution model and map work differ.

On Baseline Choice. Our nearest-site baseline uses a KD-tree. Its tree construction and query are both timed. The combined energy performs more work and should not be interpreted as a faster replacement for ordinary Voronoi when the additional controls are unnecessary.

10 LIMITATIONS AND FUTURE WORK

The current field is finite, planar and nonperiodic. Periodic tiling, curved-surface evaluation, mobile performance and cross-vendor numerical behavior require additional implementation and testing. Four preview samples do not solve minification or temporal aliasing; baked maps require suitable downstream filtering. The controls interact, and stronger phase can reduce target fidelity. Small or nearly tied cells are sensitive to precision and sampling. The square-root height has steep gradients near zero, and the site phase convention is discontinuous. The CPU fallback uses raster differences for normals and is not covered by GPU timing/diagnostic claims. Material names indicate visual inspiration; physical fracture and artist preference have not been validated. An artist study remains future work, not an established result.

11 CONCLUSION

MORONI combines polygon targets, signed-distance assignment, angular phase and coverage release as a practical cellular texture recipe. Matched experiments show that clipping contributes most of the silhouette fidelity; polygon-aware arbitration adds a smaller benefit, while phase can trade fidelity for variation. Conservative disk bounds and adaptive finite search support acceleration, with approximation error and numerical tests reported separately. Shared geometry, measured GPU execution and explicit export semantics connect the browser studio to the CPU artifact. The limitations remain part of the technique's intended scope.

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A FULL PROCEDURAL GENERATOR ALGORITHMS

To highlight the mathematical foundations, practical benefits, and drawbacks of existing cellular noise functions, we provide complete algorithmic formulations for Standard Voronoi, Power diagrams, Additively Weighted Apollonius diagrams, Worley noise, and MORONI.

A.1 Standard Voronoi Generator (L_2 Euclidean)

Standard Voronoi partitions $\mathbb{R}^2$ into convex polyhedral cells based on Euclidean distance to point sites $\mathbf{u}_c$:

$$L_{\text{Vor}}(\mathbf{x}) = \arg \min_{c \in \{0,\dots,N-1\}} \|\mathbf{x} - \mathbf{u}_c\|_2. \tag{12}$$

Algorithm 2 Standard Voronoi Generator

Require: Query point $\mathbf{x}$, sites $\{\mathbf{u}_c\}_{c=0}^{N-1}$
$d^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{0, \dots, N-1\}$ **do**
  $d \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2$
  if $d < d^\star$ **then**
    $d^\star \leftarrow d$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Nearest-site queries admit established spatial data structures; cells partition the domain and have straight edges.

Drawbacks. Site positions indirectly affect shape. Square cells are possible, but arbitrary independent polygon targets are not generally specified by sites alone. Separate clipping or distance thresholds can add gaps.

A.2 Power (Laguerre) Diagram Generator

Power diagrams assign scalar weights $w_c \in \mathbb{R}$ to each site, defining the power distance [1]:

$$\text{pow}(\mathbf{x}, c) = \|\mathbf{x} - \mathbf{u}_c\|_2^2 - w_c, \qquad L_{\text{Power}}(\mathbf{x}) = \arg\min_c \text{pow}(\mathbf{x}, c). \tag{13}$$

Algorithm 3 Power (Laguerre) Diagram Generator

Require: Query point $\mathbf{x}$, weighted sites $\{(\mathbf{u}_c, w_c)\}_{c=0}^{N-1}$
$p^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{0, \dots, N-1\}$ **do**
  $p \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2^2 - w_c$
  if $p < p^\star$ **then**
    $p^\star \leftarrow p$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Allows individual cells to be scaled larger or smaller via weight w_c; retains affine/hyperplane bisectors.

Drawbacks. Cells are strict intersections of half-planes and are mathematically guaranteed to be convex in the continuous plane; sites with small weights can be entirely eclipsed (empty cells); cannot produce concavity or fracture.

A.3 Additively Weighted (Apollonius) Diagram

Apollonius diagrams subtract weights directly from the Euclidean distance:

$$d_{\text{Apol}}(\mathbf{x}, c) = \|\mathbf{x} - \mathbf{u}_c\|_2 - w_c, \qquad L_{\text{Apol}}(\mathbf{x}) = \arg\min_c d_{\text{Apol}}(\mathbf{x}, c). \tag{14}$$

Algorithm 4 Apollonius Diagram Generator

Require: Query point $\mathbf{x}$, weighted sites $\{(\mathbf{u}_c, w_c)\}_{c=0}^{N-1}$
$d^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{0, \dots, N-1\}$ **do**
  $d \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2 - w_c$
  if $d < d^\star$ **then**
    $d^\star \leftarrow d$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Boundaries are hyperbolas, admitting curved cell boundaries and possible nonconvexity.

Drawbacks. Nonempty cells are connected in the continuous plane; site positions and additive weights jointly govern boundaries, without explicit polygon targets.

A.4 Worley Cellular Texture Basis

Worley noise [23] computes the distances $F_1 \le F_2 \le \dots$ to ordered nearest feature points:

$$F_1(\mathbf{x}) = \min_c \|\mathbf{x} - \mathbf{u}_c\|, \quad F_2(\mathbf{x}) = \min_{c \neq c^\star} \|\mathbf{x} - \mathbf{u}_c\|. \tag{15}$$

The classic cellular texture is obtained via $F_2(\mathbf{x}) - F_1(\mathbf{x})$.

Algorithm 5 Worley Cellular Basis (F_1, F_2)

Require: Query point $\mathbf{x}$, sites $\{\mathbf{u}_c\}_{c=0}^{N-1}$
$F_1 \leftarrow \infty$; $F_2 \leftarrow \infty$
for all $c \in \{0, \ldots, N-1\}$ **do**
  $d \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2$
  if $d < F_1$ **then**
    $F_2 \leftarrow F_1$; $F_1 \leftarrow d$
  else if $d < F_2$ **then**
    $F_2 \leftarrow d$
  end if
end for
return $(F_1, F_2, F_2 - F_1)$

Practical Benefits. Produces continuous intensity gradients useful for stone outlines and caustic bubbles.

Drawbacks. Underlying cell partition is identical to Voronoi; thresholds on $F_2 - F_1$ can form seams, but their physical width depends on the configuration and chosen threshold.

A.5 MORONI Generator

Algorithm 6 Full MORONI Generator with Grout Release

Require: Query point $\mathbf{x}$, cells $\{(\mathbf{u}_c, P_c, \mathrm{LUT}_c)\}$, weights α, β, γ, frequencies κ, λ, margin m
$E^\star \leftarrow \infty$; $c^\star \leftarrow 0$; $s^\star \leftarrow 0$
for all cells c in increasing index order **do**
  $\mathbf{d} \leftarrow \mathbf{x} - \mathbf{u}_c$; $d_c \leftarrow \|\mathbf{d}\|_2$
  $s_c \leftarrow$ analytical distance, or hybrid lookup with exact exterior
  $p_c \leftarrow 0$
  if $d_c \geq 10^{-6}$ **then**
    $\theta_c \leftarrow \mathrm{atan2}(d_y, d_x)$
    $p_c \leftarrow 1 - \cos(\kappa\theta_c + \lambda d_c)$
  end if
  $E \leftarrow \alpha d_c + \beta s_c + \gamma p_c$
  if $E < E^\star$ **then**
    $E^\star \leftarrow E$; $c^\star \leftarrow c$; $s^\star \leftarrow s_c$
  end if
end for
if $s^\star > m$ **then**
  return $\varnothing$ ▷ Released as grout/mortar background
else
  return $c^\star$
end if

B PRACTICAL BENEFITS VS. DRAWBACKS COMPARATIVE MATRIX

Table 7 presents an operational and mathematical comparison across all five cellular generators.

C ANALYTICAL ERROR BOUND FOR BILINEAR LUT SAMPLING

Let $s_c(\mathbf{x})$ be the exact signed distance and $\hat{s}_c$ its bilinear approximation. The lookup half-width is $B_c = 1.5r_c$ and grid spacing $h_c = 2B_c/(L-1)$. For exact corner samples, signed distance is 1-Lipschitz. At offset (a, b) within a square, the bilinear value is a convex combination of its corner values; the error is bounded by their weighted mean distance to the query. Cauchy–Schwarz gives

$$|s_c - \hat{s}_c| \leq \sqrt{a(h_c - a) + b(h_c - b)} \leq h_c/\sqrt{2} = \varepsilon_c. \tag{16}$$

Outside the box the implementation evaluates analytical distance, giving zero approximation error there. Floating-point roundoff is additional. The bilinear reconstruction need not itself be 1-Lipschitz, and switching to analytical evaluation can create a small box-boundary discontinuity. The largest bound among default polygons is 0.01639 for $L = 72$, compared with maximum observed error 0.00847 in Table 4.

With $\varepsilon = \max_c \varepsilon_c$, an analytical winning label whose lead over every competitor exceeds $2\beta\varepsilon$ is preserved under the hybrid approximation. A smaller lead need not change the label. This condition alone supplies no universal area or mismatch fraction without further regularity assumptions about competing energy fields.

D GPU FRAGMENT SHADER PIPELINE

The WebGL fragment shader computes the three-term energy from an uploaded float32 geometry texture, traversing all finite candidates with disk-bound rejection. Polygon distance uses segment projection and a ray-crossing sign test. Phase uses the direct expression, with the same explicit center convention as the CPU. The shader supports analytical distance; it does not use the CPU spatial LUT.

For retained pixels define $h = \sqrt{\mathrm{clamp}(-s_c/r, 0, 1)}$ and zero for background. Normal differences use world offset $\epsilon = .005s$:

$$\mathbf{n} = \mathrm{normalize}(-.06s\, h_x, -.06s\, h_y, 1). \tag{17}$$

RGB encodes $(\mathbf{n} + 1)/2$; positive image y points down. Consumers using the opposite tangent convention must invert green. Roughness is the illustrative constant .25 in cells and .95 in gaps. Data maps are linear data; albedo is an unlit display palette with no baked surface lighting. Exports render one sample per pixel and exclude wireframe overlays. The SVG command exports generating polygons only. These map choices are not recovered physical material properties.

E INTERACTIVE LABORATORY STUDIO SHOWCASE AND MATERIAL PRESETS

To enable immediate exploration and verification by peer reviewers, artists, and graphics practitioners, we provide an interactive laboratory studio: **MORONI Studio**. Figure 7 illustrates the complete user interface running live in a standard web browser.

E.1 Laboratory Architecture and Live Modes

The laboratory operates entirely client-side with zero dependencies or server requirements:

(1) **Interactive Dials Sidebar**: Sliders update the field and WebGL uniforms. End-to-end interaction latency is distinct from the measured GPU draw time.
(2) **Top Viewport Mode Bar**: Allows instantaneous toggling between:

Table 7: Comprehensive comparative feature matrix: Standard Voronoi, Power Diagrams, Apollonius Diagrams, Worley Noise, and MORONI.

Property	Voronoi	Power	Apollonius	Worley $F_2 - F_1$	MORONI
Primitive	Sites	Weighted sites	Weighted sites	Ordered distances	Polygons and sites
Shape control	Via sites	Via sites/weights	Via sites/weights	Via sites	Target polygons
Gap control	Extra threshold	Extra threshold	Extra threshold	Field threshold	Coverage margin
Continuous cells	Convex	Convex	Connected	Voronoi partition	No general guarantee
Boundary model	Bisectors	Affine bisectors	Hyperbolas	Distance difference	Competing energies
Acceleration	Spatial queries	Established structures	Established structures	Spatial queries	Bounded finite search
Cost per candidate	Distance	Squared distance	Distance	Distance	Polygon SDF and phase

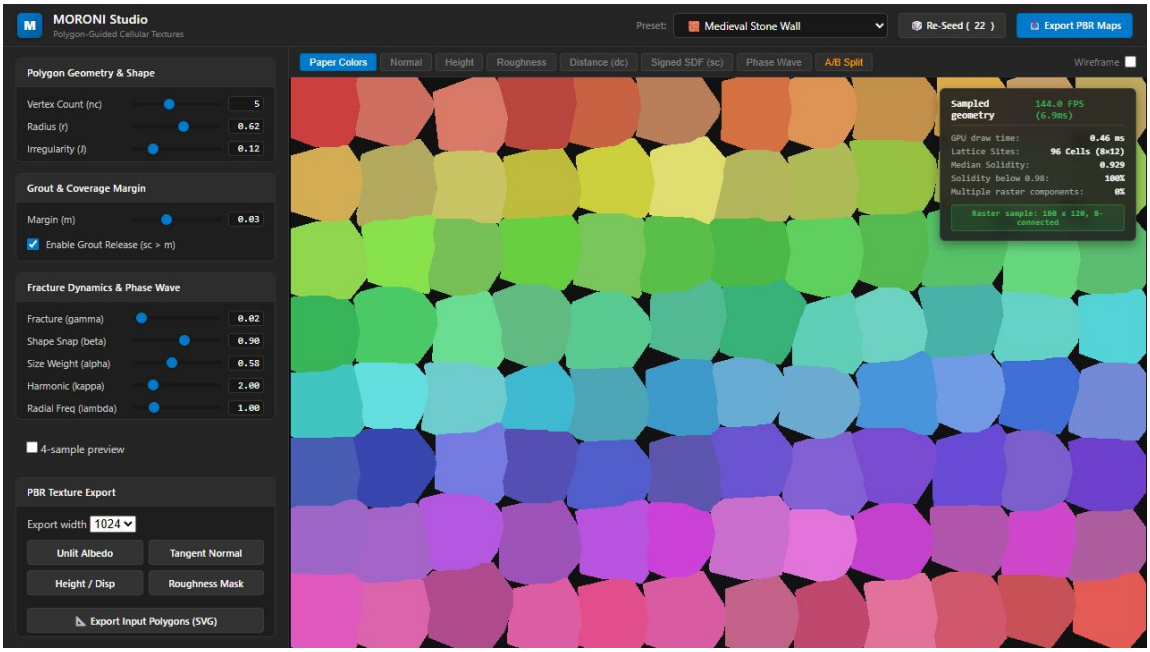


Figure 7: The MORONI Studio interactive laboratory interface running via WebGL2. Left: Interactive control sidebar exposing the five art-direction dials (cell shape n_c, size r, irregularity J, coverage margin m, and fracture parameters γ, κ, λ), seed randomization, and PBR export controls. Centre/Right: Interactive real-time viewport with top view-mode selector and upper-right live HUD reporting geometric solidity, low-solidity percentage and retained raster components. The current version uses Paper Colors and removes shaded mode and diagram-family classification.

Table 8: Parameter specifications for the 12 illustrative material presets in MORONI Studio.

Preset Name	n_c	Radius r	Jitter J	Margin m	Use Margin	α	β	γ	κ	Visual Surface Characteristics
Stone Wall	5	0.62	0.12	0.03	✓	0.58	0.90	0.02	2.0	Polygonal blocks with gaps
Cracked Mud	7	0.60	0.40	0.05	✓	0.58	0.32	0.30	2.0	Irregular regions and crack-like gaps
Ceramic Tile	4	0.60	0.02	0.01	✓	0.58	1.20	0.00	2.0	Quadrilateral targets with small coverage gaps
Shattered Glass	4	0.70	0.50	0.08	✓	0.58	0.32	0.35	3.0	Phase-distorted regions and irregular seams
Spiderweb	6	0.65	0.30	0.04	✓	0.50	0.35	0.42	4.0	Higher-harmonic angular boundary variation
Cobblestone	6	0.64	0.25	0.06	✓	0.55	0.75	0.05	2.0	Six-sided targets with coverage gaps
Sci-Fi Hex	6	0.60	0.00	0.02	✓	0.58	1.35	0.00	2.0	Regular six-sided targets with no phase modulation
Craquelure	5	0.55	0.35	0.015	✓	0.60	0.40	0.22	2.0	Five-sided targets with moderate phase variation
Dragon Scales	5	0.68	0.18	0.02	✓	0.52	0.85	0.08	1.0	Five-sided targets with first-harmonic modulation
Alien Biomass	6	0.62	0.45	0.04	✓	0.48	0.45	0.28	3.0	Irregular targets with third-harmonic modulation
Basalt Columns	6	0.60	0.08	0.012	✓	0.58	1.10	0.00	2.0	Six-sided targets with low angular jitter
Voronoi Baseline	6	0.55	0.00	0.00	×	1.00	0.00	0.00	2.0	Standard nearest-site Euclidean perpendicular bisectors

- **Paper Colors**: Unlit cell identity colors shared with the Python figures. Shaded mode and edge darkening are removed.
- **Tangent-Space Normal Map**: Dynamic 3D surface normals computed via central finite differences.

- **Height / Displacement Map**: Grayscale elevation map mapping interior signed distance to dome curvature.
- **Roughness / Grout Mask**: Two-valued mask distinguishing smooth cell tiles from rough recessed mortar.
- **Energy Field Decomposition**: Direct visual inspection of the isolated components: centre distance d_c, signed polygon field $s_c(\mathbf{x})$, and angular phase interference wave $(1 - \cos)$.
- **A/B Split-Screen Comparison**: Allows dragging an interactive divider directly across the screen to contrast standard Voronoi against MORONI.

(3) **Live Geometric & Topological HUD**: A 180×120 GPU label readback drives raster solidity and retained 8-connected component counts. The minimum component area is 10^{-4} of the sampled domain. These are descriptive samples, not a proof of diagram-family membership or continuous topology. GPU timing is marked unavailable where timer queries are unsupported.

E.2 Material Preset Specifications

Table 8 lists the parameter configurations for the 12 illustrative material presets packaged within the studio. Figure 8 demonstrates the 4-channel PBR texture map export pipeline (diffuse, normal, height, roughness), providing authoring inputs that require suitable color-space, tangent-normal, scale and filtering choices in a material renderer. The full preset frequency values λ are 1.0 except glass 1.8, spiderweb 2.5, cobblestone 1.2, craquelure 2.2, scales 1.4 and biomass 1.5.

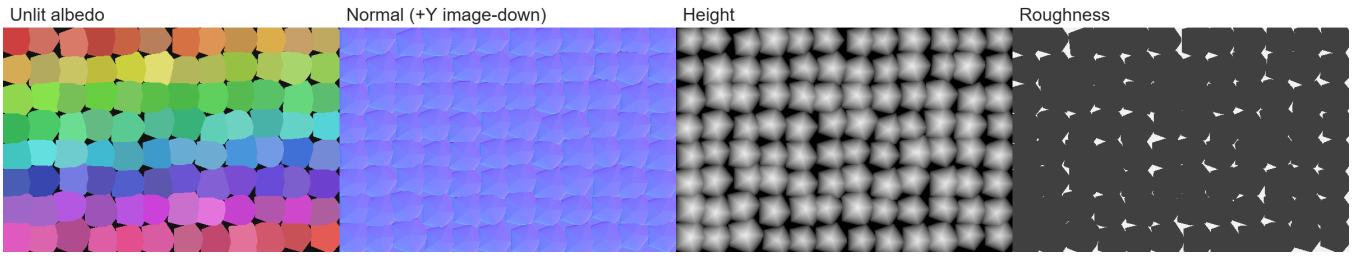


Figure 8: Four-channel PBR texture maps synthesized and exported directly from the MORONI Studio laboratory: (a) Unlit identity albedo, (b) Tangent-space normal map derived via central differences of the height field, (c) Height displacement map with dome curvature, and (d) Roughness / grout mask.